Neural Control of Respiration

Dr Shahneela Siraj

LEARNING OBJECTIVES

At the end of lecture the student should be able to

- Recall the different respiratory centers
- Describe the functions of respiratory centers & their role in eliciting the respiratory cycle
- Explain about the mechanisms of control of breathing
- Describe the effects of lesions in these centers.

Control of Respiration

- Goals of regulation of ventilation is to keep arterial levels of O_2 & CO_2 constant

- Maintain Rate of respiration
- Maintain Depth of respiration

Control of breathing can therefore be simplified to the fact that *both the rate and depth of breathing are regulated so that alveolar PCO_2 is maintained close to 40 mmHg*

Two Mechanism are present for respiratory control

- Nervous regulation
- Chemical regulation

Neural Control of Respiration

Two separate neural mechanisms regulate respiration. One is responsible for

- **Voluntary control**

and the other for

- **Automatic/Spontaneous control**



CONTROL SYSTEMS

- The voluntary control system is located in the cerebral cortex and sends impulses to the respiratory motor neurons via the corticospinal tracts.
- The automatic system is driven by a group of pacemaker cells in the medulla. Impulses from these cells activate motor neurons in the cervical and thoracic spinal cord that innervate inspiratory muscles. Those in the cervical cord activate the diaphragm via the phrenic nerves, and those in the thoracic spinal cord activate the external intercostal muscles. However, the impulses also reach the innervation of the internal intercostal muscles and other expiratory muscles.
- The motor neurons to the expiratory muscles are inhibited when those supplying the inspiratory muscles are active, and vice versa.

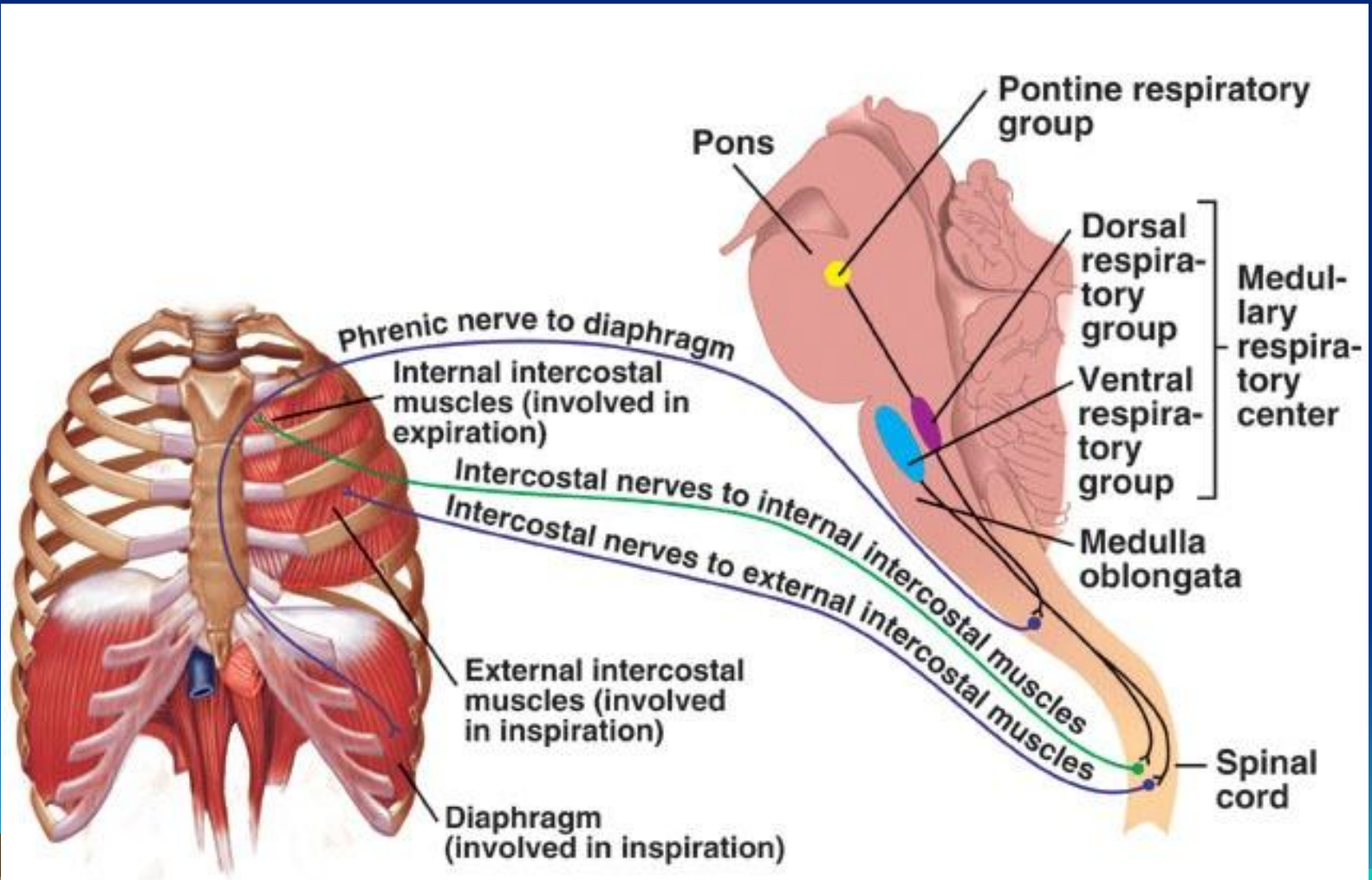
NERVOUS REGULATION

- Adjust the rate of alveolar ventilation according to the demands of body
- PO_2 and PCO_2 in the arterial blood hardly altered even during respiratory distress
- Lungs can maintain the pao_2 and $paco_2$ within the normal range, even under widely varying conditions by regulation from respiratory centre
- **Respiratory center**
  - **Those areas of the brain that stimulate the**
    - **Contraction of the diaphragm &**
    - **intercostal muscles**

REGULATION OF RESPIRATION BY CNS

- **Breathing**
  - an **involuntary ,** rhythmic act
    - **Rhythmicity** produced by orderly discharge of neurons supplying the respiratory muscles
    - initiated from the respiratory areas present in **medulla oblongata and pons**
- neurons in the **brainstem**
  - **automatic control of unconscious breathing**
- neurons in the **motor cortex** of the cerebrum
  - **voluntary control**

Respiratory Structures in Brainstem



RESPIRATORY CONTROL SYSTEM IN CNS

- Breathing is controlled by centers in the brain stem There are four components to this control system:

1- Control **centers** for breathing in the **brain stem (medulla oblongata and pons)**

2- **Chemoreceptor** for O_2 *and* CO_2

3- Mechanoreceptor in the *lungs* and *joints*

4- The **respiratory muscles**, whose activity is directed by the brain stem centers. Voluntary control can also be exerted by commands from the cerebral cortex (e.g., breath-holding or voluntary hyperventilation), which can temporarily override the brain stem.

BRAIN STEM CONTROL OF BREATHING

- Controlled by the **medulla and pons of the brain stem**
- **three groups of neurons** or **brain stem centers**
  - **The medullary rhythmicity center**
  - **The apneustic center**
  - **The pneumotaxic center**

Function

- Control Frequency **of normal, involuntary breathing**

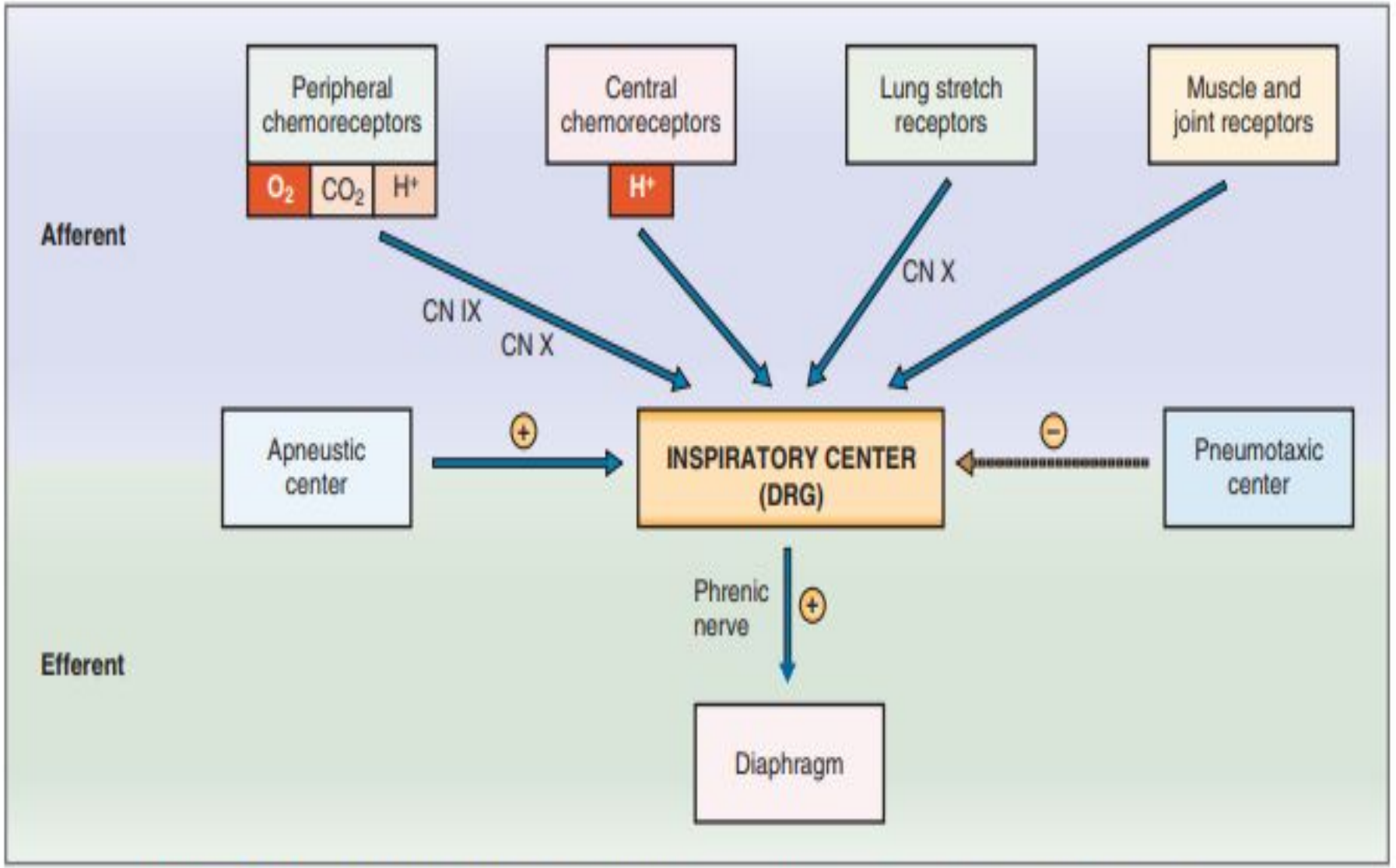


Figure 5–32 Brain stem control of breathing. Afferent (sensory) information reaches the medullary inspiratory center via central and peripheral chemoreceptors and mechanoreceptors. Efferent (motor) information is sent from the inspiratory center to the phrenic nerve, which innervates the diaphragm. CN, Cranial nerve; DRG, dorsal respiratory group.

Pre-Bötzinger Complex

- Cluster of interneurons in the ventrolateral medulla of the Brainstem
- Contains Pace maker Cells-Spontaneous Breathing
- Also called central control pattern generator b/c it plays most fundamental role in control of respiration

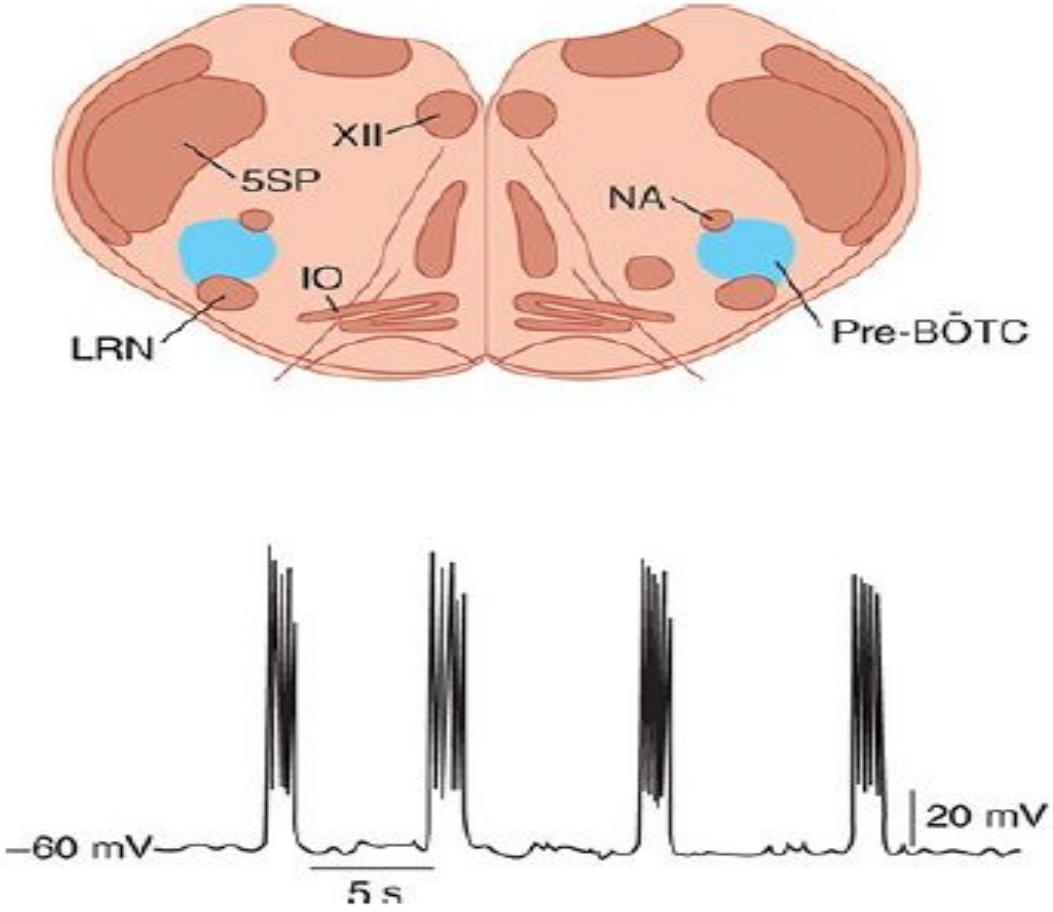
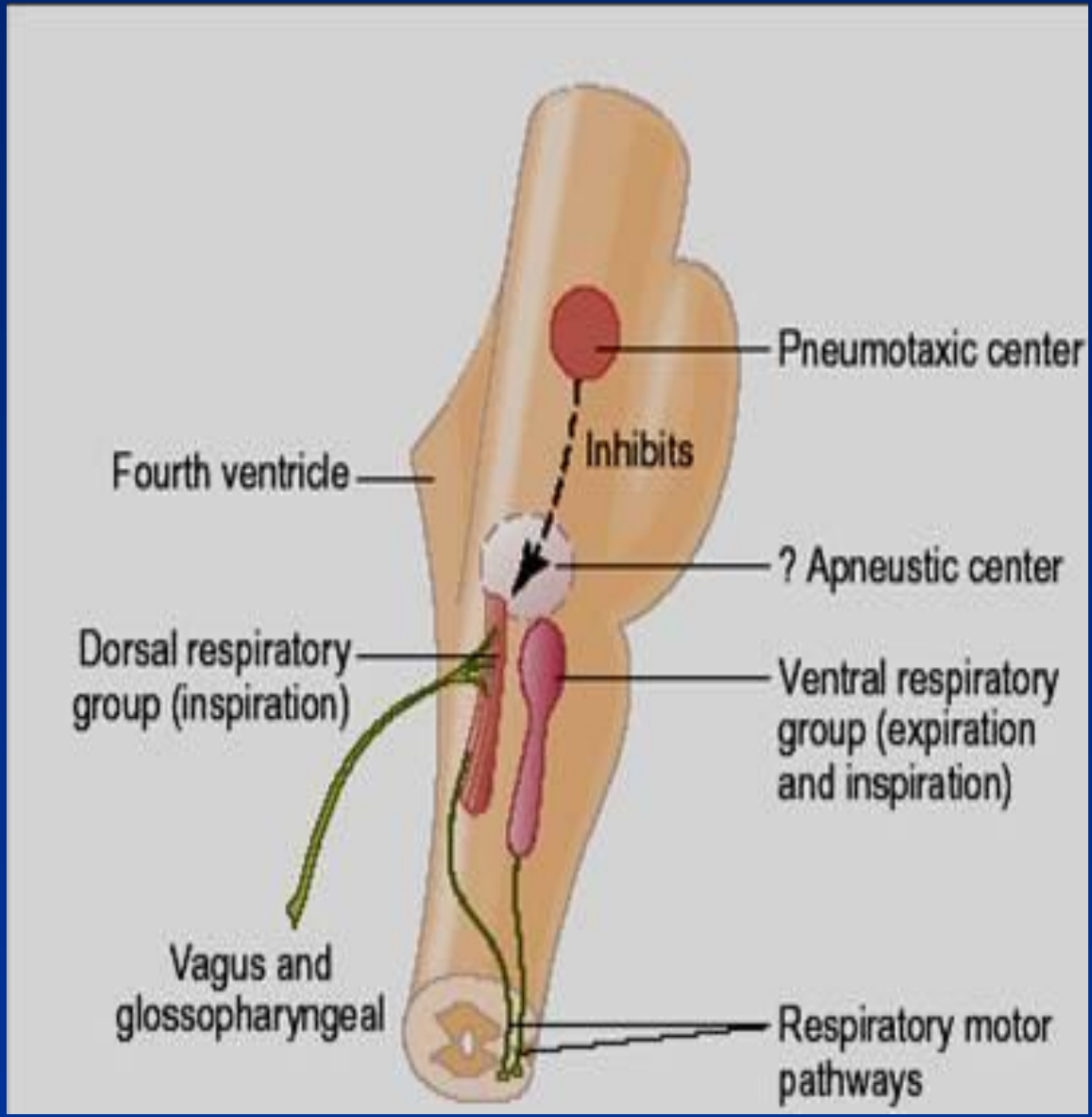


FIGURE 36–2 Pacemaker cells in the pre-Bötzinger complex (pre-BÖTC). **Top:** Anatomic diagram of the pre-BTC from a neonatal rat. **Bottom:** Sample rhythmic discharge tracing of neurons in the pre-BTC complex from a brain slice of a neonatal rat. IO, inferior olive; LRN, lateral reticular nucleus; NA, nucleus ambiguus; XII, nucleus of 12th cranial nerve; 5SP, spinal nucleus of trigeminal nerve. (Modified with permission from Feldman JC, Gray PA: Sighs and gasps in a dish. Nat Neurosci 2000;3(6):531–532.)

Neural control of Respiration



Rate and depth of respiration is regulated by certain groups of neurons , lies bilaterally in reticular formation of medulla and pons. These groups of neurons collectively known as Respiratory centers.

- **Four different neuronal groups of respiratory centre ?**

1. Dorsal respiratory group or inspiratory area
2. Ventral respiratory group or expiratory area
3. Pneumotaxic area
4. Apneustic area

1. DRG (dorsal respiratory group) or inspiratory centre

- **It lies bilaterally in the reticular formation of the medulla on dorsal portion and interconnected with the nucleus of the tractus solitarius (NTS).**
- **The NTS receive the sensory inputs from the vagus and glossopharyngeal nerves.**
- These CN IX & X receives input from peripheral chemoreceptors.

This centre Primarily concerned with inspiration.

- **On stimulation**

  they send their impulses to diaphragm & inspiratory muscles, and due to contraction of these muscles inspiration take place for about 2 seconds.
- **After this inspiratory wave**

  the activity of inspiratory centre is inhibited so insp. muscles relaxed and expiration take place.

- So dorsal group is the main driving force for inspiration.

- How is rhythm generated?
  - There is a continuous inspiratory drive from the DRG and inspiration take place for about 2 seconds then an intermittent inhibitory signal that inhibits inspiration and causes expiration.

Inspiratory "Ramp" Signals

- Rhythmical Inspiratory discharge from DRG
- Nervous signal that is transmitted to the inspiratory muscles, mainly the diaphragm
- Begins weakly & increases steadily for about 2 secs in normal respiration (inspiration)
- Then ceases abruptly for approx next 3 secs (expiration)

Control of Inspiratory Ramp

1. Control of the rate of increase of the ramp signal rate of ramp signal - ramp increases rapidly - fills the lungs rapidly
2. Control of the limiting point at which the ramp suddenly ceases

- Usual method for controlling the rate of respiration; shortens the duration of Inspiration & Expiration (frequency of respiration)

Contd…

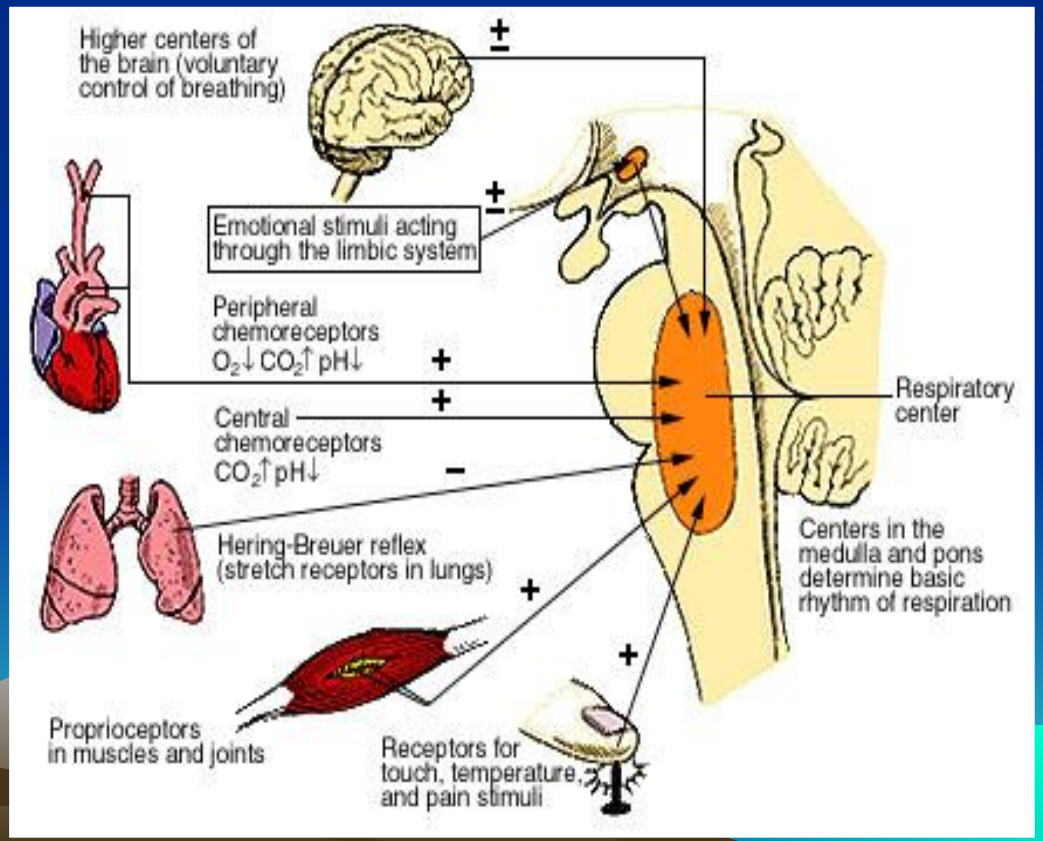
- Stopping the Ramp
- Shortens the rate of inspiration and expiration
- Increasing the frequency of respiration

INPUT TO INSPIRATORY CENTER

- Receives **sensory input** mainly from:
  - **Peripheral chemoreceptor via**
  - **Glossopharyngeal nerve and vagus nerve**
  - **Mechanoreceptors in the lungs via vagus nerve.**

Also receive information from:

- **Stretch receptors** in the lungs via cranial nerve x.
- From **peripheral chemoreceptor** in the region where cranial nerve ix and x leave the brainstem.
- From **ph and pco2 receptors** in the bone-dura-arachnoid csf space.

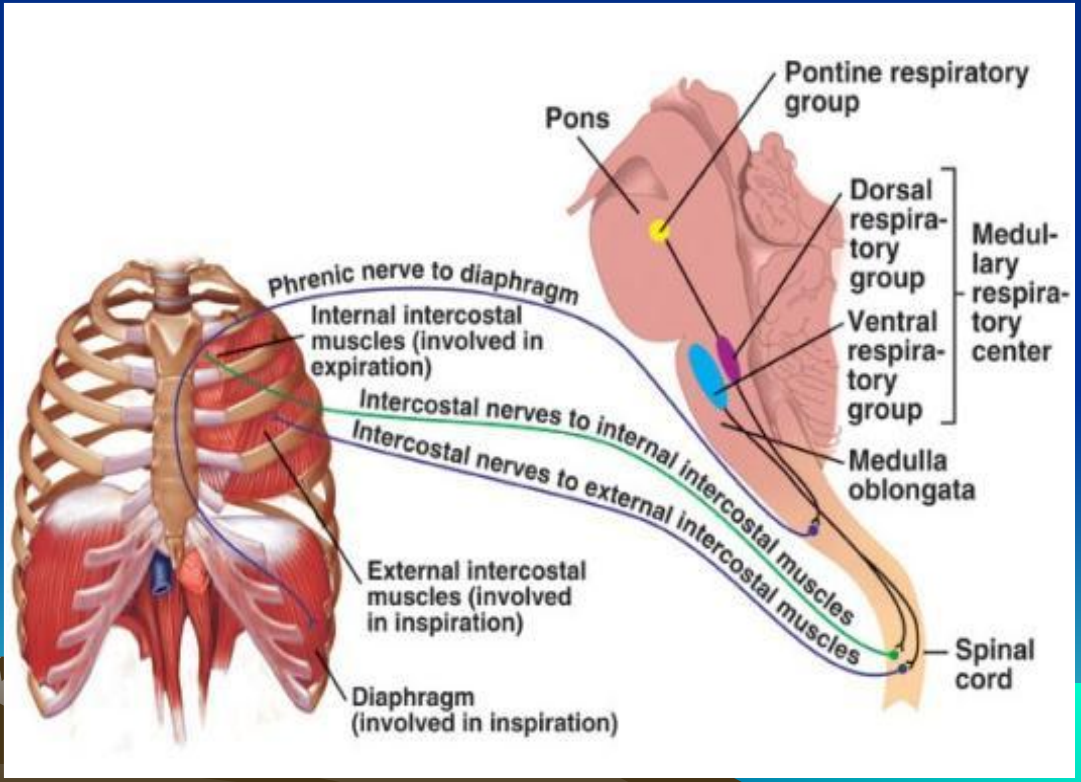


EFFERENTS FROM INSPIRATORY CENTER

- Efferent fibers go to **spinal cord** synapse with **lower motor neuron** in cervical and thoracic region
- Efferents from spinal cord
  - **phrenic nerves** to the diaphragm
  - **intercostal nerves** to the intercostal muscles.

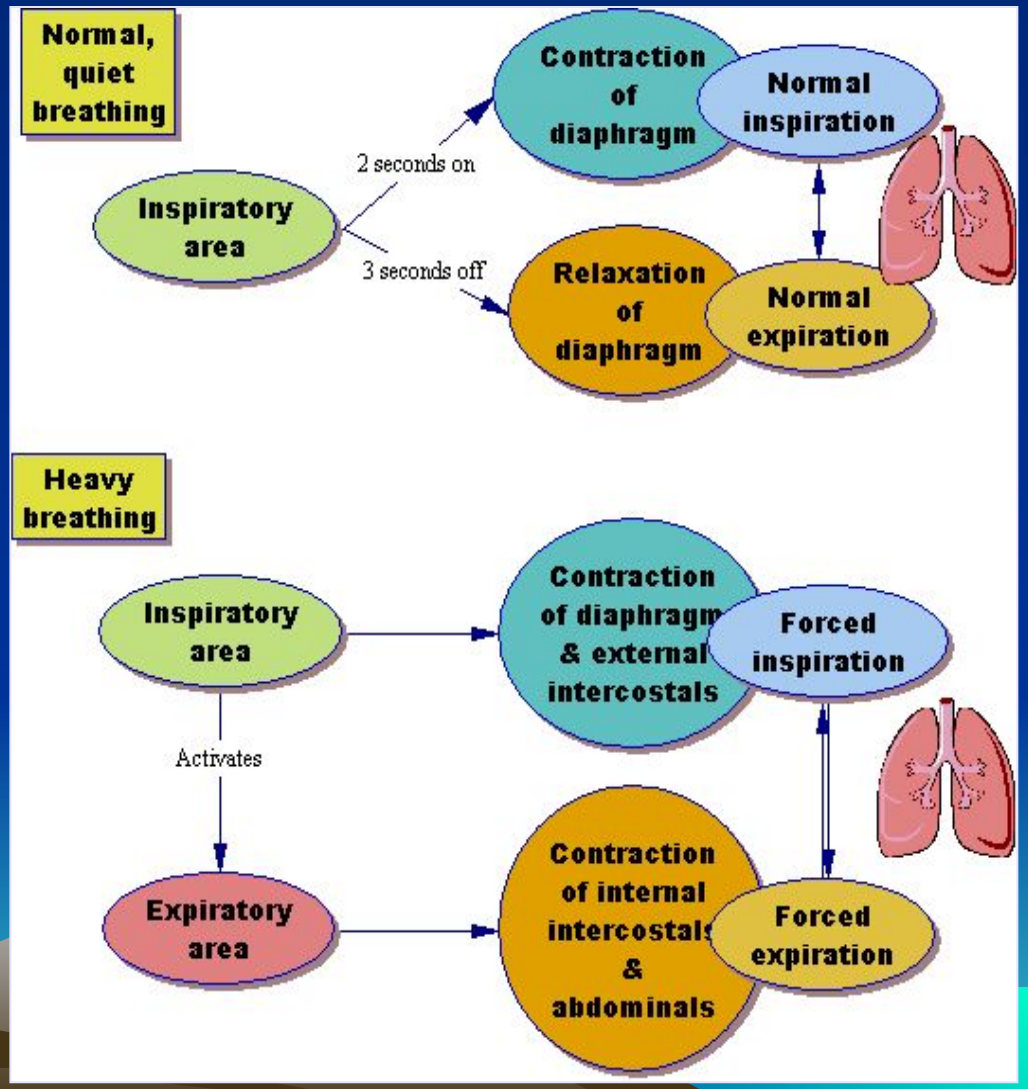
Final Response:

- **contraction** of diaphragm and intercostal muscle
- inspiration occur.

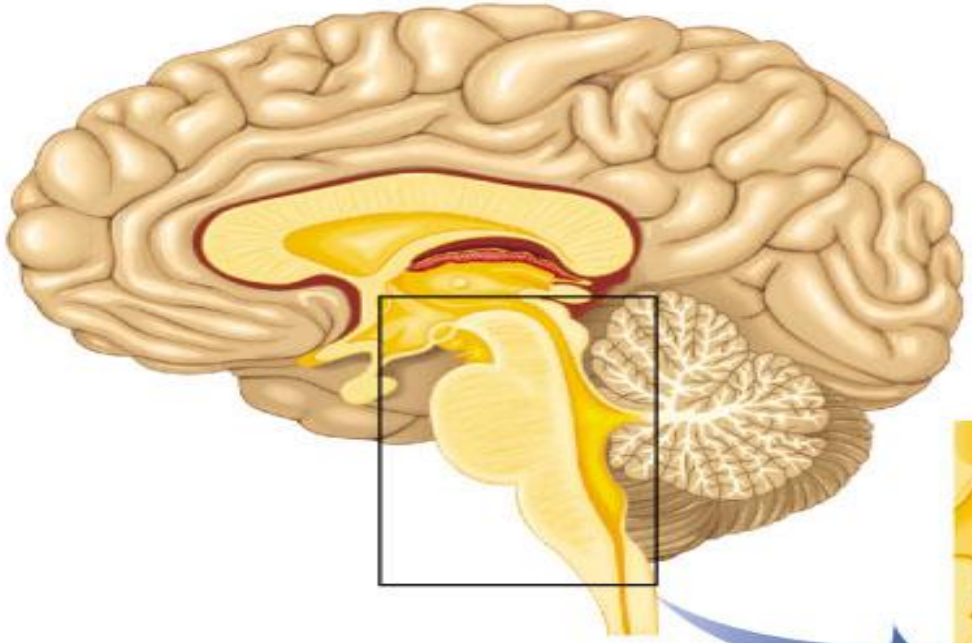


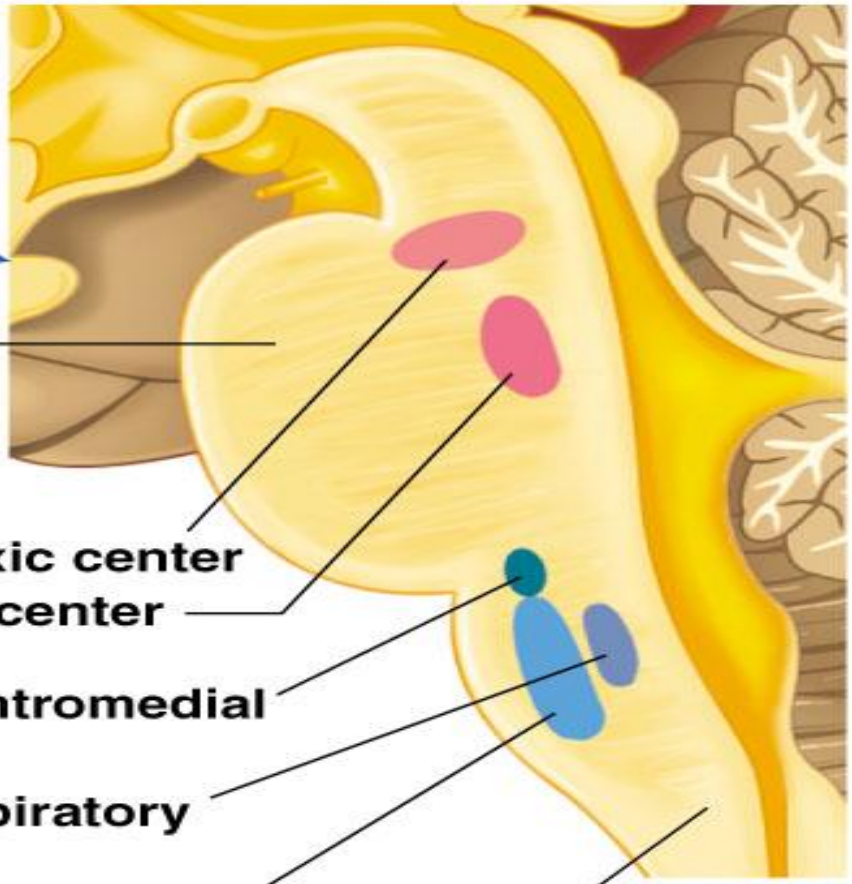
FUNCTION OF INSPIRATORY CENTER

- **Establish the basic rhythm of breathing**
- During quiet breathing
  - **inspiration last for 2 seconds**
  - **expiration last for 3 sec.**



© 2001 Brooks/Cole - Thomson Learning





Respiratory control centers in brain stem

- **Pons respiratory centers**
  - Pneumotaxic center
  - Apneustic center
- **Medullary respiratory center**
  - Rostral ventromedial medulla
  - Dorsal respiratory group
  - Ventral respiratory group

2. VRG (ventral respiratory group) Or expiratory area

lies bilaterally in the reticular formation of the medulla on ventral portion in the nucleus retroambiguus.

- Normally they remain silent during normal breathing.
- They become active during deep expiration ,send signal to expiratory muscles ,which depress the rib cage ,for deep expiration.

- It can stimulate both inspiratory & expiratory respiratory muscles during increased ventilatory drive.

Pneumotaxic center:

located dorsally in N. parabrachialis of upper **pons.**

- inhibits the duration of inspiration by turning off DRG ramp signal after start of inspiration,
- so after 2 seconds of inspiration, this centre send its inhibitory impulses to DRG ,to inhibit inspiration so expiration take place.

So it causes

….. decrease in length of respiration

….. increase rate of respiration

- Strong activity: short breaths
- Weak activity: long breaths

- When the pneumotaxic center is inactivated, inspiration is prolonged (apneusis).
- Damage to Pneumotaxic center : Results in increased depth of inspiration

Apneustic Center:

- **located in the lower part of the pons. (at the junction of pons and medulla)**
- functions to prevent inhibition of DRG under some circumstances
- **Sends signals to the DRG that prevent the off-switch for inspiration-*causes sustained inspiration (apneusis)***
- **Probably works in conjunction with the pneumotaxic center to control the depth of inspiration**

So it causes

- …..increase in length of respiration
- …..decrease rate of respiration

Mechanism of Rhythmic Ventilation

- Medullary respiratory center neurons are continuously active (spontaneous/automatic breathing)
- Combined input from all sources (receptors, brain) causes action potentials to stimulate respiratory muscles

To Increase Inspiration

More and more neurons get activated

To Stop Inspiration

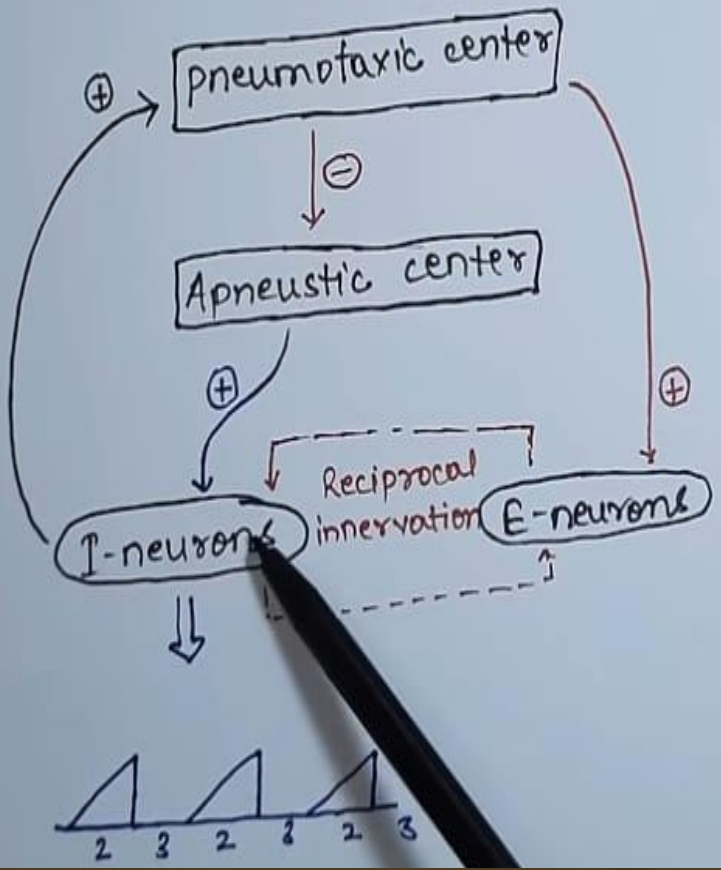
Neurons receive input from pontine group and stretch receptors in lungs.

Inhibitory neurons activated because Inspiratory switch-off.

Start of expiration

Effect: Relaxation of Inspiratory muscles

DRG	VRG
Inspiratory ramp	Forced breathing
Pneum. center "switch off" of Inspiratory Ramp	Apneustic center prevents "switch-off"




THANK YOU!